Neutral Current Pion Production at MINERvA

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Abstract

This report presents a study of neutral current (NC) coherent π^0 production using data from the MINERvA experiment. In NC coherent pion production, a neutrino scatters off an entire nucleus producing a single forward-going neutral pion without breaking the nucleus. This offers a clean test of the partially conserved axial current (PCAC) hypothesis in the low momentum transfer limit and plays a critical role in neutrino oscillation experiments by contributing to backgrounds in appearance channels. In addition to coherent π^0 production, we investigate diffractive π^0 production, where the scattering happens on a proton and produce similar final states.

1 Neutrino Physics

The Standard Model is a quantum field theory that describes 17 fundamental particles, along with their antiparticles and the forces that govern their interactions. Of these 17 particles, 12 are fermions (particles with half-integer spin), and the remaining five are bosons (particles with integer spin). The Higgs boson is responsible for generating mass for other particles, while the other bosons mediate the three fundamental forces – electromagnetic, strong nuclear, and weak interactions. The fermions are divided into two categories known as leptons and quarks. Quarks, which carry electric charges of +2/3e or +1/3e, are always found in groups and form composite particles such as mesons ($q\bar{q}$) and baryons (qqq or $\bar{q}\bar{q}\bar{q}$). Leptons consist of three charged particles – electrons, muons, and tauons – as well as three neutral leptons, the neutrinos, which are of particular interest in my research.

Neutrinos are electrically neutral fermions that interact only via the weak force (and gravity), making them uniquely valuable probes of weak-scale physics. Despite early predictions from the Standard Model that they were massless, we now know neutrinos have small but nonzero masses – at least six orders of magnitude smaller than that of the electron. Their weak interaction cross-sections are correspondingly tiny, allowing them to traverse vast amounts of matter with minimal scattering. This exceptionally low interaction rate enables high-momentum-transfer weak processes to occur with minimal background from softer interactions, making neutrinos clean probes of quantum interference phenomena such as flavor oscillations. The sensitivity of oscillation patterns to mass-squared differences and mixing angles provides a direct window into physics beyond the Standard Model, including mass generation mechanisms, CP violation, and possible sterile neutrino states.

We can detect a neutrino in our experiment only if it interacts with matter. Neutrinos interact with matter solely via the weak interaction, which occurs in two ways: charged-current (CC) and neutral-current (NC) channels. In CC interactions, the neutrino converts into its equivalent charged lepton (e.g. inverse beta decay, $\nu_e + p \rightarrow n + e^+$) and interactions occur through the exchange of $W^\pm$ bosons. The experiment detects the charged lepton. In contrast, in NC interactions, the neutrino remains a neutrino but transfers energy and momentum to whatever it interacted with through the exchange of a Z boson. We detect this energy transfer, either because the target recoils (e.g., neutrino-electron scattering, $\nu + e \rightarrow \nu + e$) or because it breaks up (e.g., ${}^2H + \nu \rightarrow p + n + \nu$).

Various detector technologies have been used in neutrino experiments over the years, depending on the requirements of the particular study. Thus, neutrino experiments require low energy thresholds, good angular and time resolution, accurate energy measurement, and better particle and charge identification properties.

The various neutrino detection technologies include radiochemical detectors (HOMESTAKE, SAGE), liquid scintillator technologies (Borexino, MiniBooNE), tracking detectors (MINOS, MINERvA, ICARUS), Water Cherenkov detectors (Super Kamiokande, IMB) and many more. Tracking detectors are good at distinguishing between different event topologies and reconstructing events containing multiple particles (e.g., $\nu_\mu + p \rightarrow \mu + n + N\pi$, where $N \geq 1$). MINERvA uses a scintillator tracker for studying neutrino interactions.

2 MINERvA Experiment and NUMI Beamline

Neutrinos at the Main Injector (NuMI) at Fermilab provides the neutrino beam for MINERvA detector. NuMI

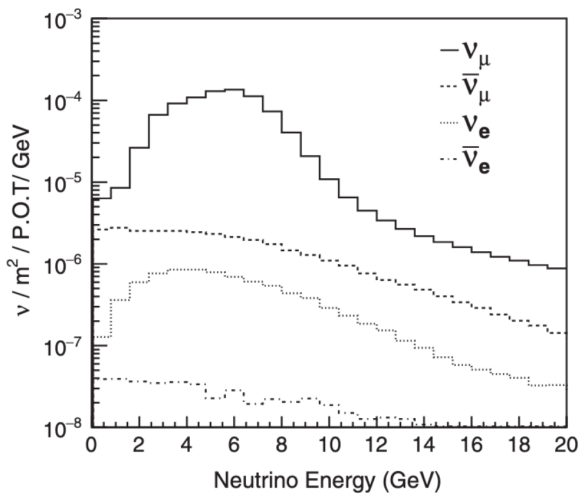


Figure 1: Simulated neutrino fluxes at MINERvA versus neutrino energy in the ME neutrino-mode configuration of the NuMI beam.

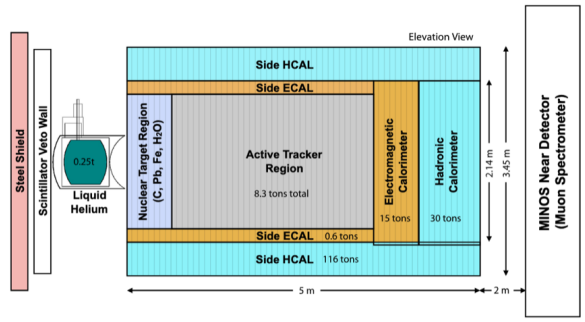


Figure 2: Schematic of MINERvA detector

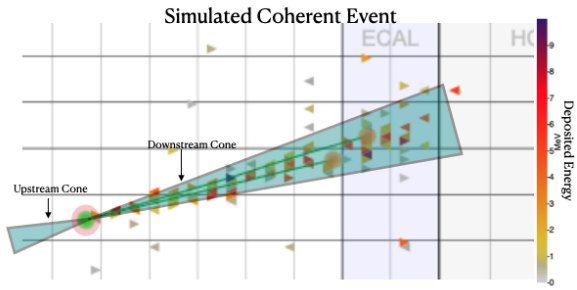


Figure 3: Event display of a simulated neutral current coherent event in the detector. The downstream cone and upstream cone are also shown. The two photons from the π^0 decay are contained in the downstream cone.

facility comprises a 120 GeV primary proton beam, a two interaction length graphite target, two parabolic focusing horns, and a 675 m decay pipe. Throughout MINERvA's operation, the NuMI had two configurations Low Energy (LE, 3 GeV) and Medium Energy (ME, 6 GeV) [1][2]. The data I am using for the cross-section analyses discussed later are from the ME configuration. The current in the focusing horns can be configured to focus positively or negatively charged particles, resulting in neutrino (FHC) or antineutrino (RHC) enhanced beams, respectively. The dataset used in my analysis corresponds to 1.11×10^{21} POT (Protons on Target) for FHC and 1.05×10^{21} POT for RHC. NuMI is simulated using a Geant4-based model of the NuMI beam line [3]. The neutrino flux predictions for the NuMI ME neutrino-mode configuration are shown in Figure 1.

The MINERvA experiment investigates neutrino interactions with various nuclei to deepen our understanding of weak interactions in nuclear environments and A-dependence in neutrino interactions. By measuring cross-sections across different nuclear targets, MINERvA helps improve models of neutrino-nucleus scattering and reduces systematic uncertainties in oscillation experiments.

A schematic of the detector is shown in Fig. 2. The MINERvA detector consists of an inner detector which contains an upstream nuclear target followed by a downstream tracker region, and an outer detector composed of electromagnetic (ECAL) and hadronic (HCAL) calorimeters. The inner detector was a hexagonal region with nuclear targets interleaved with plastic scintillator planes to track charged particles. The tracker region is made of 120 scintillator planes which are rotated by 60° with respect to adjacent ones, enabling three-dimensional reconstruction. The ECAL surrounds the inner detector, and the HCAL surrounds the ECAL. The former (latter) consists of planes of lead (iron) and scintillator to contain and track electromagnetically (strongly) interacting particles. Neutrino interactions are modeled using the GENIE event generator [4], while GEANT4 simulates final-state particle interactions and detector responses [3].

3 Coherent and Diffractive Pion Production

Coherent π^0 production, represented by the processes $\bar{\nu}_\mu A \rightarrow \mu^+ A \pi^-$ for CC and $\bar{\nu} A \rightarrow \bar{\nu} A \pi^0$ for NC interactions, is a weak interaction phenomenon where the incident neutrino scatters off an entire nucleus A producing a pion while leaving the nucleus intact and in its ground state via exchange of a charged $W^\pm$ boson or a neutral Z boson respectively. In coherent scattering, no quantum numbers are transferred to the target nucleus. All the allowed channels for the interaction are depicted in Figure 4.

The wavelength of the momentum transferred by the meson to the nucleus is larger than the nucleus radius,

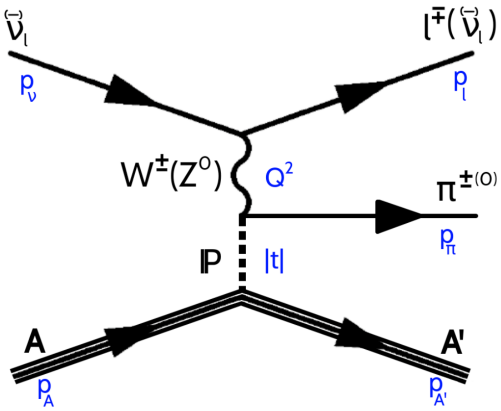


Figure 4: Feynman diagram of the neutrino-induced inelastic coherent pion production. Black letters denote particles, while blue letters represent their four-momenta. According to the PCAC approach, the π-meson transfers momentum to the nucleus by the exchange of a pomeron represented by $\mathbb{P}$. The diagram is taken from [5].

making the protons and neutrons in the nucleus react in phase (coherently) as a whole, and leaving the nucleus in its initial state, with just a small recoil energy. For the interaction to remain coherent, the square of the four momentum transferred to the nucleus, $|t| = |(p_\nu - p_l - p_\pi)^2|$ or $|p_A - p_{A'}|^2$, should be very small, with the minimum momentum given by [6]

$$|t|_{\text{min}} = \left(\frac{Q^2 + m_\pi^2}{2E_\pi}\right)^2 \tag{1}$$

where Q^2 is the square of the four-momentum carried by the $W^\pm$ or Z boson, m_π and E_π are the mass and energy of the pion respectively. The interaction probability falls off exponentially with $|t|$, and the coherence threshold is [7]

$$|t|_{\text{coh}} = \frac{(\hbar c)^2}{R_N^2} \tag{2}$$

where c is the speed of light, $\hbar$ is the reduced Planck constant, and R_N is the nuclear radius.

Further, similar interactions where the neutrino scatters off a proton are termed as diffractive processes. In these processes, the proton recoils to a direction opposite to the neutrino beam to conserve momentum. The negative of the four momentum transfer to the target proton, $-t = |t|$ since $t \leq 0$, is related to the final state kinetic energy of the proton by $T_p = |t|/2M_p$, where M_p is the target proton mass.

In case of NC coherent or diffractive processes, the inability to observe the final-state lepton prevents the calculation of $|t|$. Thus, NC processes are mainly identified by the decay of produced π^0 into two photons, and subsequent electromagnetic showers due to e^+e^- pair production.

3.1 Motivation

NC Coherent π^0 production is a significant background in long-baseline $\nu_\mu \to \nu_e$ oscillation measurements around 1 GeV [8]. Since the two decay photons produce electromagnetic showers similar to those from electrons, they can be misidentified as ν_e charged-current events, especially when one shower is undetected or the two overlap. Accurate modeling of this process helps constrain this background.

The simplest theoretical approach for describing coherent pion production is based on Adler's Partially Conserved Axial Current (PCAC) theorem, which relates the weak charged and neutral currents to the pion through the divergence of the axial-vector current [9]. This allows us to relate the neutrino induced coherent pion production to the pion-nucleus elastic scattering. Experimental simulations like GENIE generally rely on the Rein–Sehgal (RS) model [10], which is based on the PCAC hypothesis to predict neutrino induced coherent pion production. Thus, this analysis provides a way to study the weak axial vector current, by testing the predictions of PCAC framework.

NC coherent pion production is also important because it is a predicted BSM process, involving new, weakly interacting neutral particles. For example, such a particle might interact coherently with a nucleus via a NC–like process but lack a CC analog because of mass requirements to be produced at the relevant energy scale. References for such predictions can be found in [11] [12].

The cross section quantifies the likelihood of an interaction between an incoming particle and a target, expressed as an effective area. It can be thought of as the product of the physical area seen by the particle and the probability that a specific reaction will occur upon interaction. The flux-averaged cross section is defined as

$$\sigma = \frac{U^{-1}(N_{\text{data}} - N_{\text{bkg}})}{\epsilon \phi N_{\text{target}}} \tag{3}$$

where U is the matrix that relates reconstructed variables to true variables (Migration), N_{data} and N_{bkg} are the number of selected data and simulation-predicted background events, respectively, ϵ is the efficiency of the NC coherent signal selection calculated from simulation, N_{target} is the number of target nuclei in the detector fiducial volume, and ϕ is the integrated neutrino flux. The analysis characterizes the NC coherent π^0 production cross section as a function of emitted pion energy.

3.2 Signal and Sidebands

To isolate the interaction of interest from the full dataset, we apply a set of selection cuts on reconstructed detector variables. Events that pass these criteria form our analysis sample.

A signal event is one that genuinely arises from the interaction under study, regardless of whether it is correctly identified. However, similar final-state signatures can result from different processes, and some detector responses may be ambiguous or incomplete. These misidentified events contribute to the background, which can mimic the signal despite not originating from the targeted interaction. Even after selection on reconstructed variables, some background events survive the selection. To quantify both signal and background contributions in our region of interest, we use truth-level information from Monte Carlo (MC) simulations.

In this analysis, we define sideband regions in reconstructed space dominated by background, to study how background events behave in the selected region. These sidebands are thus essential for validating and tuning our background models, improving our ability to isolate the signal in the primary analysis region.

The identification of NC coherent π^0 production events is based on the selection methodology described in Ref. [13]. The analysis is performed in the tracker region of the MINERvA detector, and the data used correspond to the ME neutrino flux from the NuMI beamline, which peaks at approximately 6 GeV. Muons from charged current (CC) ν_μ interactions typically leave long tracks that exit the detector. Hence events are selected by requiring no tracks exiting the detector and having at least one reconstructed track originating within the fiducial volume. The remaining tracks for these events are then evaluated to identify electromagnetic shower-like behavior, which is indicative of $\pi^0 \to \gamma\gamma$ decays. For each reconstructed track, an electromagnetic cone object is constructed to contain the potential shower, as shown in Figure 3. The cone is defined as the union of a cylinder of radius 50 mm extending from the event vertex (origin of track) along the track direction and a 7.5° cone with an apex at the event vertex and a symmetry axis along the track direction. All energy deposits within this downstream cone region are clustered and sorted by their distance from the vertex. These clusters are evaluated based on their radiation lengths – mean distance in a material over which the energy of an electron is reduced by the factor 1/e – to distinguish EM showers from other particle types. If a gap of 3 or more radiation lengths exists between neighboring clusters, all clusters from that gap onward are excluded from the cone. The total energy within the cone (E_{el}) is computed, and particle identification (PID) requirements, as described in [13], are applied to assess consistency with a π^0 signal.

In addition to these criteria, the cone object is required to have its vertex within the fiducial volume of the detector. To avoid backgrounds from hadronic interactions, clusters with more than six reconstructed tracks originating from the start point of the EM candidate track are eliminated. Fake signal events that arise from photomultiplier afterpulsing are also eliminated.

Further discrimination is achieved using energy deposition characteristics. Specifically, the longitudinal energy profile is used to calculate the mean energy loss per unit length (dE/dx), and the energy deposited outside the EM cone is defined as the available energy (E_{avail}). An additional variable, the Inline Upstream Energy (IUE), quantifies the energy deposited within a 7.5° cone directed upstream along the axis opposite the primary EM cone. The energy signature of the recoil proton in diffractive processes appears within this upstream cone. These variables provide further rejection of background processes and contribute to the isolation of a clean NC coherent π^0 sample [14] [15].

This analysis aims to study events with $E_{\pi^0} > 1.5$ GeV. This ensures other interactions with the same initial and final states, such as $\nu p \to \nu\Delta^+$ where $\Delta^+ \to p\pi^0$, do not significantly interfere with pion production. This requirement translates to a requirement of $E_{\mathrm{el}} > 1.5$ GeV for signal events. The signature of NC coherent π^0 production in MINERvA would appear as two forward tracks originating from a single vertex with no additional energy deposition near the vertex. Thus the signal region needs signature of two pairs of e^+e^- and a mean $dE/dX > 2.4$ MeV/cm. To discriminate events where nucleus recoil and constitutes the diffractive background, events with $IUE < 10$ MeV are selected. Finally to identify events where the nucleus is not broken apart, we select events with $E_{\mathrm{avail}} < 0.5$ GeV.

There are three major background contributions that can mimic this signal: incoherent π^0 production (NCPi0, CCPi0), where the final state includes a neutral pion but the nucleus is broken apart; single electron neutrino CC events, where the outgoing electron initiates an electromagnetic shower similar to that from a π^0 decay (CCQE-like, not CCQElike); and diffractive events with higher nuclear recoil (NCDiff).

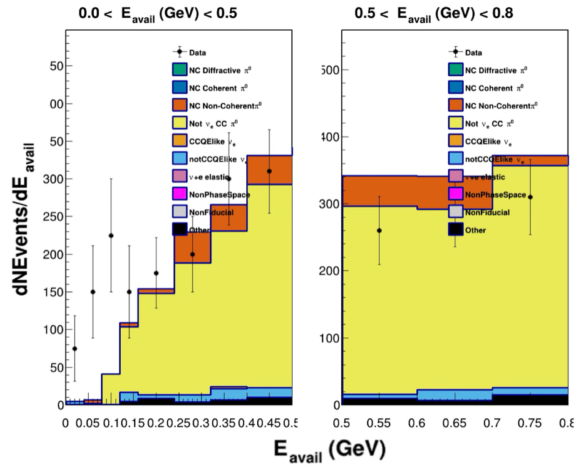
Three sidebands – Pi0, Electron neutrino, and Diffractive – are constructed, each enriched in a specific type of background. The Pi0 sideband is defined by an available energy in the range 0.5 GeV $< E_{\mathrm{avail}} <$ 0.8 GeV, the electron neutrino sideband is characterized by $dE/dx < 2.4$ MeV/cm, and the diffractive sideband is identified by an $IUE > 10$ MeV at least 5 cm (two planes of scintillator strips) upstream of the electromagnetic candidate. These sidebands serve as control regions dominated by specific background processes, enabling us to study and constrain their contributions in the signal region through comparisons between data and Monte Carlo predictions.

As discussed, incoherent NC and CC events with a π^0

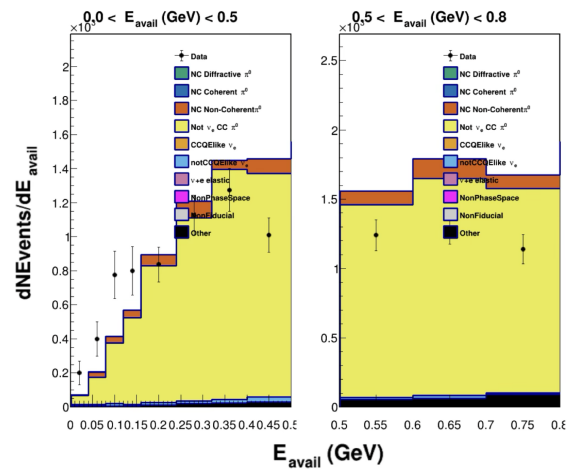
in the final state (NCPi0 and CCPi0, respectively) constitute important background processes in this analysis. These interactions are typically deep inelastic scattering (DIS) events, where the neutrino interacts at the quark level, resulting in the production of mesons or baryons. These backgrounds are studied using the Pi0 sideband and help assess how accurately MINERvA models the pion background. To isolate a sample dominated by incoherent π^0 interactions, we look for events in our selection that have a reconstructed track extending into the side ECAL and the outer detector (OD). These tracks are typically classified by the reconstruction as muons due to their topological features, but in this context, they are more likely to originate from π^0 interactions misidentified as muons. The corresponding pure π^0 selection is shown in Figure 5. The E_{avail} distribution in this sideband is then compared between data and Monte Carlo (MC) simulation. If the background is modeled correctly, the data-to-MC (Data/MC) ratio should be consistent between the signal region and the Pi0 sideband.

From the calculation this ratio is 0.931 ± 0.061 in the signal region and 0.730 ± 0.050 in the Pi0 sideband region in the FHC mode, and 1.017 ± 0.151 in the signal region and 0.815 ± 0.118 in the Pi0 region in RHC configuration. This indicates an underestimation in Pi0 background prediction by around 30 percent in the signal, diffractive, and electron neutrino sidebands. Further in order to verify whether this discrepancy has a dependence on E_{el}, the double ratio – $(Data/MC)_{\text{signal}}/(Data/MC)_{\text{sideband}}$ – in each bins of E_{el} is studied. This study confirmed that the data/MC disparity is consistent across all bins of E_{el}. In order to address this, the model needs to be corrected by a constant factor of 1.261 ± 0.143 (average of the ratio of Data/MC in sideband to signal in FHC and RHC). This weight is applied to NCPi0 and CCPi0 backgrounds in regions other than Pi0 sideband.

The preliminary selection for the signal and sideband region are shown in Figure 6.



(a) RHC



(b) FHC

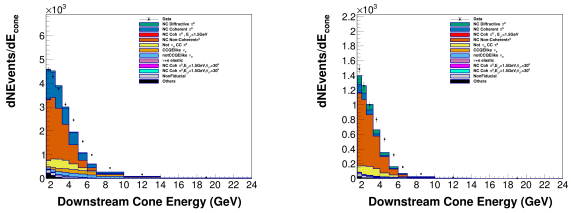
Figure 5: Bin width normalized E_{avail} distributions.

3.3 Reweighting and Background Subtraction

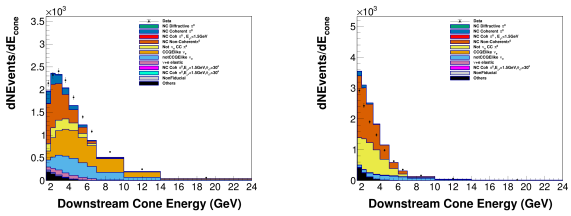
The selected event sample includes not only signal events but also background events, which must be accounted for. To constrain these backgrounds, we use the sideband regions to compare data with Monte Carlo (MC) predictions. Discrepancies between data and simulation in each of these regions are quantified by solving a linear equation for all i, $\sum_j MC_{ij} w_j = Data_i$. Here, MC_{ij} is the number of simulated background events of type j in region i, w_j is a scaling weight to be applied to that background type, and $Data_i$ is the number of observed events in that region. These weights are then applied to the corresponding background components in the signal region, allowing us to subtract the appropriately scaled background and isolate the signal. The weighted distributions are shown in Figure 7.

3.4 Migration matrix and Unfolding

The data obtained are in terms of energies observed by the detector, and during reconstruction, some events may be assigned to the wrong true energy bin. We need to correct for these effects of detector smearing and relate the observed energies to the truth energies of neutral pion. The simulation prediction of how the reconstructed energies relate to the true energies is visualized through the migration matrix. A row-normalized migration matrix as shown in Figure 8, predicts the probability that events in each true or pion energy bin are reconstructed into different reconstructed or electron cone energy bin. This matrix will be close to diagonal if the detector had no smearing. The inverse of this migration matrix yields

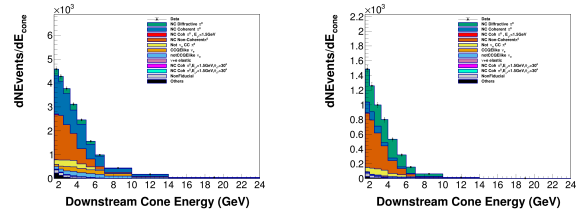


(a) Selection (b) Diffractive sideband

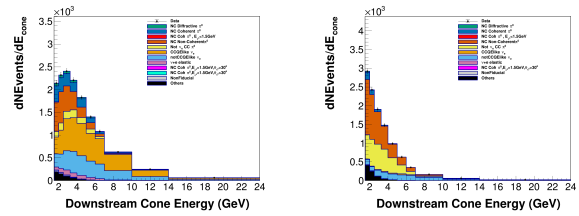


(c) Electron Neutrino sideband (d) Pi0 sideband

Figure 6: Preliminary binwidth-normalized E_{el} distribution for selection and sidebands for FHC.



(a) Selection (b) Diffractive sideband



(c) Electron Neutrino sideband (d) Pi0 sideband

Figure 7: Binwidth-normalized weighted Eel distribution for selection and sidebands.

the unsmearing matrix, $U_{j\alpha}$.

In this analysis, the inverses of these singular or poorly conditioned matrices are obtained using the pseudo-inverse method via Singular Value Decomposition (SVD). This approach regularizes the inversion and mitigates numerical instabilities caused by singular values close to zero. The number of singular values to discard is determined by performing chi-squared tests and examining the correlation matrices.

The true pion energies ($E_{\pi^0_j}$) can be recovered by solving $(E_{\pi^0})_j = \sum_j U_{j\alpha} \cdot (E_{\mathrm{el}})_\alpha$.

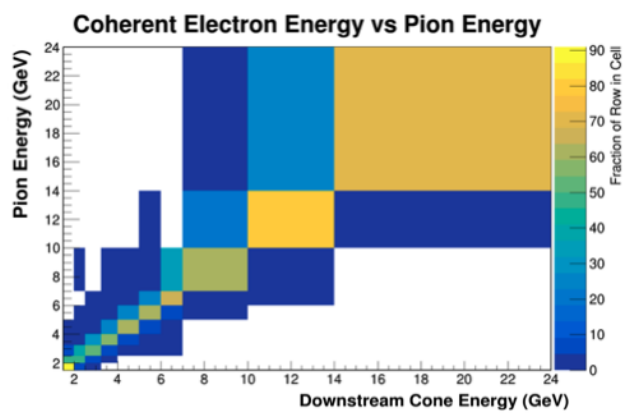


Figure 8: Migration matrix for E_{π^0} vs E_{el}.

3.5 Efficiency correction and Cross-section Results

Neutrino event signatures are complicated and difficult to identify. Our selection cuts to remove background events will also cut out some fraction of the signal. The efficiency– number of simulated events that pass the signal selection– is shown as a function of pion energy in Figure 9. A reduction in efficiency at low pion energy is obtained because the reconstruction method assumes the two photons are coalesced into a single electromagnetic cone object, which is less likely at lower energies.

The cross section for the discussed NC coherent pion production can be measured from the true pion energy distribution by correcting for efficiency and normalizing using MINERvA's predicted neutrino flux, constrained by *in situ* measurements and the number of carbon atoms in the detector [16]. The cross sections for FHC and RHC modes are obtained as Figure 10.

3.6 Systematic Uncertainties

Systematic uncertainties in the cross-section measurement stem from limitations in modeling the neutrino flux, detector response, and neutrino interaction physics, all of which rely heavily on MC simulations. The flux model predicts the energy spectrum and composition of the neutrino beam, using hadron production data, e.g., data from the NA49 experiment [17]. The detector model simulates how particles deposit energy, are tracked, and are reconstructed, incorporating effects like energy calibration, material composition, and resolution. The interaction model, primarily implemented using GENIE, describes how neutrinos interact with nuclei, including quasi-elastic scattering, resonance production, and final state interactions (FSI). To estimate the impact

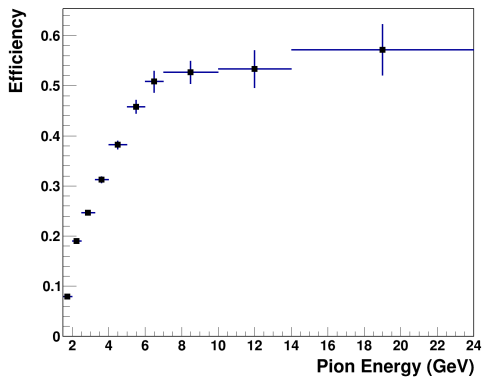


Figure 9: Selection efficiency for each bin of E_{π^0}

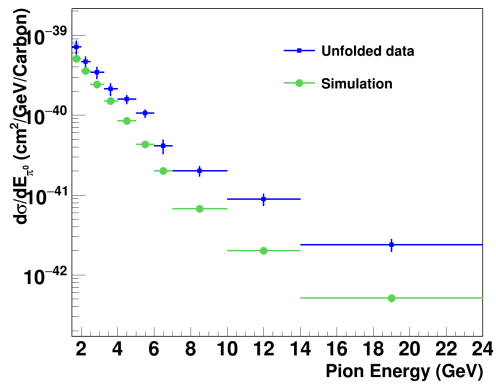
of uncertainties, MINERvA employs a many-universe method, where model parameters are randomly varied within known uncertainties to generate alternative simulated outcomes. Each variation (or universe) is compared to the central value (CV) prediction to measure its effect on the final result. Systematic shifts can be lateral, changing the variables associated with the event, or vertical, altering the probability that this specific event occurs. These uncertainties are grouped into categories – flux, detector, interaction, and MnvTunes – and are propagated through all analysis steps to determine their contribution to the total uncertainty [13]. The breakdown of systematic uncertainty on the measured cross section is shown in Figure 11.

3.7 Four Momentum Transfer Study

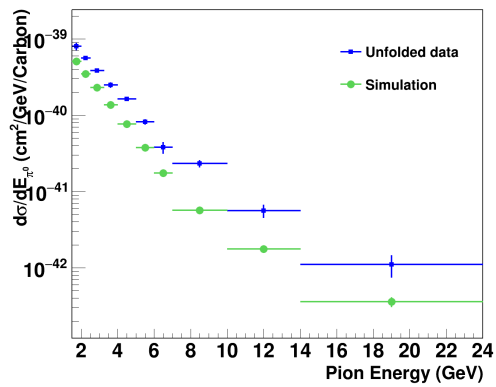
In NC coherent interactions, the final-state neutrino is not observed, and the observed final state is a lone π^0. As a consequence, neither energy of the neutrino nor $|t|$ can be directly measured. Since $|t|$ is not available, experiments isolate NC coherent interactions using the coherence condition [18]

$$\mathcal{L} = E_\pi(1 - \cos\theta_\pi) \lesssim \frac{\hbar c}{\sqrt[3]{A}(1.12\ fm)} \tag{4}$$

where E_π is the pion energy, θ_π is the angle between beam direction and π^0 meson, and A is the atomic number. The denominator on the right-hand side corresponds to charge radius of the nuclei (R). In our selection we are examining this distribution and tuned distributions for the coherent and diffractive regions are shown in Figure 12. The shape of the acceptance-corrected distribution can be fit to an exponential form $e^{-R\mathcal{L}/\hbar c}$, to extract the coherent radius of carbon and hydrogen nuclei.



(a) FHC



(b) RHC

Figure 10: NC coherent pion production cross section on log scale.

4 Neutral Current Diffractive Pion Production

Diffractive π^0 production is a background to charged current ν_e interactions and to $\nu + e^- \to \nu + e^-$ elastic scattering. Therefore, it is important to quantify its rate and determine its characteristics. It is also a background to searches for new particle which decay into, or diffractively produce electromagnetic final states. When the analysis was earlier carried out by the collaboration, the model correction to the incoherent Pi0 backgrounds was not considered during the selection and requires a re-evaluation. This analysis measures the cross section as function of pion energy and t using the procedures discussed in the previous section and the work is ongoing.

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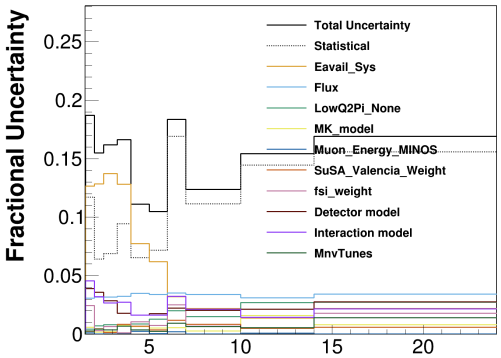
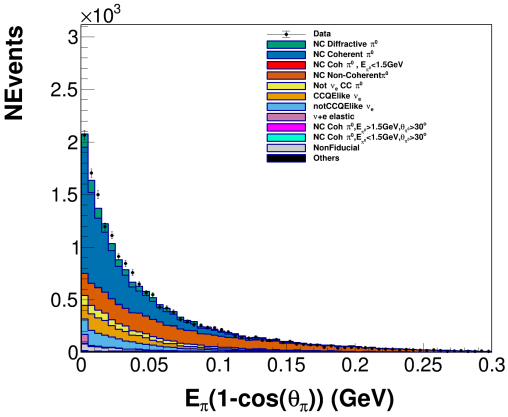
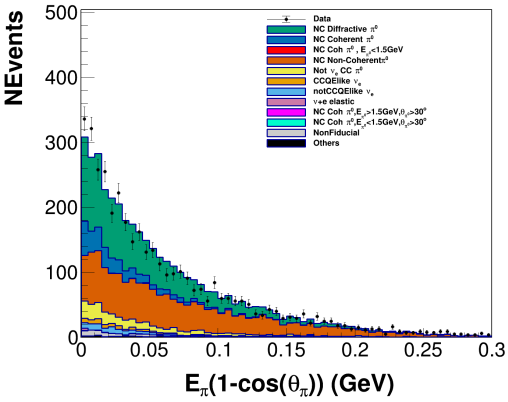


Figure 11: Systematic uncertainty breakdown on cross section in FHC.



(a) Coherent



(b) Diffractive

Figure 12: Tuned distributions of measure of four momentum transfer as discussed in Eqn. 4 for the coherent (top) and diffractive (bottom) π^0 samples.

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